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Elon Musk: Tesla, SpaceX, and the Quest for a Fantastic Future

by Ashlee Vance

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91 Highlights

Highlight (Yellow) | Location 343

To my great surprise, Musk cut me off after a couple of minutes and simply said, “Okay.” One thing that Musk holds in the highest regard is resolve, and he respects people who continue on after being told no. Dozens of other journalists had asked him to help with a book before, but I’d been the only annoying asshole who continued on after Musk’s initial rejection, and he seemed to like that.

Highlight (Yellow) | Location 359

Turning humans into space colonizers is his stated life’s purpose. “I would like to die thinking that humanity has a bright future,” he said. “If we can solve sustainable energy and be well on our way to becoming a multiplanetary species with a self-sustaining civilization on another planet—to cope with a worst-case scenario happening and extinguishing human consciousness—then,” and here he paused for a moment, “I think that would be really good.”

Highlight (Yellow) | Location 365

Musk’s ready willingness to tackle impossible things has turned him into a deity in Silicon Valley, where fellow CEOs like Page speak of him in reverential awe, and budding entrepreneurs strive “to be like Elon” just as they had been striving in years past to mimic Steve Jobs.

Highlight (Yellow) | Location 395

Musk had built an honest-to-God rocket factory in the middle of Los Angeles.

Highlight (Yellow) | Location 467

“The best minds of my generation are thinking about how to make people click ads,” Jeff Hammerbacher, an early Facebook engineer, told me. “That sucks.”

Highlight (Yellow) | Location 505

The space business requires dealing with a mess of politics, back-scratching, and protectionism that undermines the fundamentals of capitalism. Steve Jobs faced similar forces when he went up against the recording industry

to bring the iPod and iTunes to market. The crotchety Luddites in the music industry were a pleasure to deal with compared to Musk's foes who build weapons and countries for a living.

Highlight (Yellow) | Location 508

SpaceX has been testing reusable rockets that can carry payloads to space and land back on Earth, on their launchpads, with precision. If the company can perfect this technology, it will deal a devastating blow to all of its competitors and almost assuredly push some mainstays of the rocket industry out of business while establishing the United States as the world leader for taking cargo and humans to space. It's a threat that Musk figures has earned him plenty of fierce enemies. "The list of people that would not mind if I was gone is growing," Musk said. "My family fears that the Russians will assassinate me."

Highlight (Yellow) | Location 512

With Tesla Motors, Musk has tried to revamp the way cars are manufactured and sold, while building out a worldwide fuel distribution network at the same time. Instead of hybrids, which in Musk lingo are suboptimal compromises, Tesla strives to make all-electric cars that people lust after and that push the limits of technology. Tesla does not sell these cars through dealers; it sells them on the Web and in Apple-like galleries located in high-end shopping centers. Tesla also does not anticipate making lots of money from servicing its vehicles, since electric cars do not require the oil changes and other maintenance procedures of traditional cars. The direct sales model embraced by Tesla stands as a major affront to car dealers used to haggling with their customers and making their profits from exorbitant maintenance fees. Tesla's recharging stations now run alongside many of the major highways in the United States, Europe, and Asia and can add hundreds of miles of oomph back to a car in about twenty minutes. These so-called supercharging stations are solar-powered, and Tesla owners pay nothing to refuel. While much of America's infrastructure decays, Musk is building a futuristic end-to-end transportation system that would allow the United States to leapfrog the rest of the world. Musk's vision, and, of late, execution seem to combine the best of Henry Ford and John D. Rockefeller.

Highlight (Yellow) | Location 531

When Musk sets unrealistic goals, verbally abuses employees, and works them to the bone, it's understood to be —on some level—part of the Mars agenda. Some employees love him for this. Others loathe him but remain oddly loyal out of respect for his drive and mission. What Musk has developed that so many of the entrepreneurs in Silicon Valley lack is a meaningful worldview. He's the possessed genius on the grandest quest anyone has ever concocted. He's less a CEO chasing riches than a general marshaling troops to secure victory. Where Mark Zuckerberg wants to help you share baby photos, Musk wants to . . . well . . . save the human race from self-imposed or accidental annihilation. The life that Musk has created to manage all of these endeavors is preposterous. A typical week starts at his mansion in Bel Air. On Monday, he works the entire day at SpaceX. On Tuesday, he begins at SpaceX, then hops onto his jet and flies to Silicon Valley. He spends a couple of days working at Tesla, which has its offices in Palo Alto and factory in Fremont. Musk does not own a home in Northern California and ends up staying at the luxe Rosewood hotel or at friends' houses. To arrange the stays with friends, Musk's assistant will send an e-mail asking, "Room for one?" and if the friend says, "Yes," Musk turns up at the door late at night. Most often he stays in a guest room, but he's also been known to crash on the couch after winding down with some video games. Then it's back to Los Angeles and SpaceX on Thursday. He

shares custody of his five young boys—twins and triplets—with his ex-wife, Justine, and has them four days a week. Each year, Musk tabulates the amount of flight time he endures per week to help him get a sense of just how out of hand things are getting. Asked how he survives this schedule, Musk said, “I had a tough childhood, so maybe that was helpful.”

Highlight (Yellow) | Location 554

“I think the time allocated to the businesses and the kids is going fine,” Musk said. “I would like to allocate more time to dating, though. I need to find a girlfriend. That’s why I need to carve out just a little more time. I think maybe even another five to ten—how much time does a woman want a week? Maybe ten hours? That’s kind of the minimum? I don’t know.”

Highlight (Yellow) | Location 596

Musk has taken industries like aerospace and automotive that America seemed to have given up on and recast them as something new and fantastic. At the heart of this transformation are Musk’s skills as a software maker and his ability to apply them to machines. He’s merged atoms and bits in ways that few people thought possible, and the results have been spectacular.

Highlight (Yellow) | Location 599

For the moment, he’s still making rich people’s toys, and his budding empire could be an exploded rocket or massive Tesla recall away from collapse. On the other hand, Musk’s companies have already accomplished far more than his loudest detractors thought possible, and the promise of what’s to come has to leave hardened types feeling optimistic during their weaker moments.

Highlight (Yellow) | Location 602

“To me, Elon is the shining example of how Silicon Valley might be able to reinvent itself and be more relevant than chasing these quick IPOs and focusing on getting incremental products out,” said Edward Jung, a famed software engineer and inventor. “Those things are important, but they are not enough. We need to look at different models of how to do things that are longer term in nature and where the technology is more integrated.” The integration mentioned by Jung—the harmonious melding of software, electronics, advanced materials, and computing horsepower—appears to be Musk’s gift. Squint ever so slightly, and it looks like Musk could be using his skills to pave the way toward an age of astonishing machines and science fiction dreams made manifest.

Highlight (Yellow) | Location 611

Because of Musk, Americans could wake up in ten years with the most modern highway in the world: a transit system run by thousands of solar-powered charging stations and traversed by electric cars. By that time, SpaceX may well be sending up rockets every day, taking people and things to dozens of habitats and making preparations for longer treks to Mars. These advances are simultaneously difficult to fathom and seemingly inevitable if Musk can simply buy enough time to make them work. As his ex-wife, Justine, put it, “He does what he wants, and he is relentless about it. It’s Elon’s world, and the rest of us live in it.”

Highlight (Yellow) | Location 619

THE PUBLIC FIRST MET ELON REEVE MUSK IN 1984. The South African trade publication PC and Office Technology published the source code to a video game Musk had designed. Called Blastar, the science-fiction-inspired space game required 167 lines of instructions to run. This was back in the day when early computer users were required to type out commands to make their machines do much of anything. In that context, Musk's game did not shine as a marvel of computer science but it certainly surpassed what most twelve-year-olds were kicking out at the time. Its coverage in the magazine netted Musk five hundred dollars and provided some early hints about his character. The Blastar spread on page 69 of the magazine shows that the young man wanted to go by the sci-fi-author-sounding name E. R. Musk and that he already had visions of grand conquests dancing in his head. The brief explainer states, "In this game you have to destroy an alien space freighter, which is carrying deadly Hydrogen Bombs and Status Beam Machines. This game makes good use of sprites and animation, and in this sense makes the listing worth reading." (As of this writing, not even the Internet knows what "status beam machines" are.) A boy fantasizing about space and battles between good and evil is anything but amazing. A boy who takes these fantasies seriously is more remarkable. Such was the case with the young Elon Musk. By the middle of his teenage years, Musk had blended fantasy and reality to the point that they were hard to separate in his mind. Musk came to see man's fate in the universe as a personal obligation. If that meant pursuing cleaner energy technology or building spaceships to extend the human species's reach, then so be it. Musk would find a way to make these things happen. "Maybe I read too many comics as a kid," Musk said. "In the comics, it always seems like they are trying to save the world. It seemed like one should try to make the world a better place because the inverse makes no sense."

Highlight (Yellow) | Location 634

At around age fourteen, Musk had a full-on existential crisis. He tried to deal with it like many gifted adolescents do, turning to religious and philosophical texts. Musk sampled a handful of ideologies and then ended up more or less back where he had started, embracing the sci-fi lessons found in one of the most influential books in his life: *The Hitchhiker's Guide to the Galaxy*, by Douglas Adams. "He points out that one of the really tough things is figuring out what questions to ask," Musk said. "Once you figure out the question, then the answer is relatively easy. I came to the conclusion that really we should aspire to increase the scope and scale of human consciousness in order to better understand what questions to ask." The teenage Musk then arrived at his ultralogical mission statement. "The only thing that makes sense to do is strive for greater collective enlightenment," he said. It's

Highlight (Yellow) | Location 746

Over time, Musk has ended up thinking that his brain has the equivalent of a graphics chip. It allows him to see things out in the world, replicate them in his mind, and imagine how they might change or behave when interacting with other objects. "For images and numbers, I can process their interrelationships and algorithmic relationships," Musk said. "Acceleration, momentum, kinetic energy—how those sorts of things will be affected by objects comes through very vividly."

Highlight (Yellow) | Location 762

At the dinner table, Tosca would wonder aloud about the distance from Earth to the Moon. Elon would spit out the exact measurement at perigee and apogee.

Highlight (Yellow) | Location 826

He felt awed right away—"It was like, 'Whoa. Holy shit!'"—by this machine that could be programmed to do a person's bidding. "I had to have that and then hounded my father to get the computer," Musk said. Soon he owned a Commodore VIC-20, a popular home machine that went on sale in 1980. Elon's computer arrived with five kilobytes of memory and a workbook on the BASIC programming language. "It was supposed to take like six months to get through all the lessons," Elon said. "I just got super OCD on it and stayed up for three days with no sleep and did the entire thing. It seemed like the most super-compelling thing I had ever seen." Despite being an engineer, Musk's father was something of a Luddite and dismissive of the machine. Elon recounted that "he said it was just for games and that you'd never be able to do real engineering on it. I just said, 'Whatever.'" While

Highlight (Yellow) | Location 836

"It is remarkable how many things you can get to explode," Elon said. "Saltpeter, sulfur, and charcoal are the basic ingredients for gunpowder, and then if you combine a strong acid with a strong alkaline, that will generally release a lot of energy. Granulated chlorine with brake fluid—that's quite impressive. I'm lucky I have all my fingers."

Highlight (Yellow) | Location 886

In another nod to the future, Elon and Kimbal were chatting during a class break outdoors when Wood interrupted them and asked what they were going on about. "They said, 'We are talking about whether there is a need for branch banking in the financial industry and whether we will move to paperless banking.' I remember thinking that was such an absurd comment to make.

Highlight (Yellow) | Location 897

As Musk saw it, "I just look at it as 'What grades do I need to get where I want to go?' There were compulsory subjects like Afrikaans, and I just didn't see the point of learning that. It seemed ridiculous. I'd get a passing grade and that was fine. Things like physics and computers—I got the highest grade you can get in those. There needs to be a reason for a grade. I'd rather play video games, write software, and read books than try and get an A if there's no point in getting an A. I can remember failing subjects in like fourth and fifth grade. Then, my mother's boyfriend told me I'd be held back if I didn't pass. I didn't actually know you had to pass the subjects to move to the next grade. I got the best grades in class after that."

Highlight (Yellow) | Location 954

Not long after their initial meeting, Elon invited Peter Nicholson's daughter Christie to his birthday party. Christie showed up at Maye's Toronto apartment with a jar of homemade lemon curd in hand and was greeted

by Elon and about fifteen other people. Elon had never met Christie before, but he went right up to her and led her to a couch. “Then, I believe the second sentence out of his mouth was ‘I think a lot about electric cars,’” Christie said. “And then he turned to me and said, ‘Do you think about electric cars?’”

Highlight (Yellow) | Location 960

With her angular features and blond hair, Christie fit Musk’s type, and the two stayed in touch during Musk’s time in Canada. They never really dated, but Christie found Musk interesting enough to have lengthy conversations with him on the phone. “One night he told me, ‘If there was a way that I could not eat, so I could work more, I would not eat. I wish there was a way to get nutrients without sitting down for a meal.’ The enormity of his work ethic at that age and his intensity jumped out. It seemed like one of the more unusual things I had ever heard.”

Highlight (Yellow) | Location 983

Musk responded with his usual show of force. “He would call very insistently,” she said. “You always knew it was Elon because the phone would never stop ringing. The man does not take no for an answer. You can’t blow him off. I do think of him as the Terminator. He locks his gaze on to something and says, ‘It shall be mine.’ Bit by bit, he won me over.”

Highlight (Yellow) | Location 1002

After one economics exam, Musk, Farooq, and some other students in class came back to the dorms and began comparing notes to try to ascertain how well they did on the test. It soon became clear that Musk had a firmer grasp on the material than anyone else. “This was a group of fairly high achievers, and Elon stood way outside of the bell curve,” Farooq said.

Highlight (Yellow) | Location 1037

at Penn. In December 1994, he had to come up with a business plan for one of his classes and ended up writing a paper titled “The Importance of Being Solar.” The document started with a bit of Musk’s wry sense of humor. At the top of the page, he wrote: “The sun will come out tomorrow. . . .”—Little Orphan Annie on the subject of renewable energy. The paper went on to predict a rise in solar power technology based on materials improvements and the construction of large-scale solar plants. Musk delved deeply into how solar cells work and the various compounds that can make them more efficient. He concluded the paper with a drawing of the “power station of the future.” It depicted a pair of giant solar arrays in space—each four kilometers in width—sending their juice down to Earth via microwave beams to a receiving antenna with a seven-kilometer diameter. Musk received a 98 on what his professor deemed a “very interesting and well written paper.”

Highlight (Yellow) | Location 1060

As he tells it, he would daydream at Queen’s and Penn and usually end up with the same conclusion: he viewed the Internet, renewable energy, and space as the three areas that would undergo significant change in the years to come and as the markets where he could make a big impact. He vowed to pursue projects in all three.

Highlight (Yellow) | Location 1064

Musk's insistence on explaining the early origins of his passion for electric cars, solar energy, and rockets can come off as insecure. It feels as if Musk is trying to shape his life story in a forced way. But for Musk, the distinction between stumbling into something and having intent is important. Musk has long wanted the world to know that he's different from the run-of-the-mill entrepreneur in Silicon Valley. He wasn't just sniffing out trends, and he wasn't consumed by the idea of getting rich. He's been in pursuit of a master plan all along. "I really was thinking about this stuff in college," he said. "It is not some invented story after the fact. I don't want to seem like a Johnny-come-lately or that I'm chasing a fad or just being opportunistic. I'm not an investor. I like to make technologies real that I think are important for the future and useful in some sort of way."

Highlight (Yellow) | Location 1268

Zip2 acquired a trademark for its "We Power the Press" slogan and the influx of cash kept Zip2 growing fast.

Highlight (Yellow) | Location 1298

after he had time to reflect on the Zip2 situation, Musk realized that he could have handled some of the situations with employees better. "I had never really run a team of any sort before," Musk said. "I'd never been a sports captain or a captain of anything or managed a single person. I had to think, Okay, what are the things that affect how a team functions. The first obvious assumption would be that other people will behave like you. But that's not true. Even if they would like to behave like you, they don't necessarily have all the assumptions or information that you have in your mind. So, if I know a certain set of things, and I talk to a replica of myself but only communicate half the information, you can't expect that the replica would come to the same conclusion. You have to put yourself in a position where you say, 'Well, how would this sound to them, knowing what they know?'"

Highlight (Yellow) | Location 1308

There was one guy who wrote a quantum mechanics equation, a quantum probability on the board, and he got it wrong. I'm like, 'How can you write that?' Then I corrected it for him. He hated me after that. Eventually, I realized, Okay, I might have fixed that thing but now I've made the person unproductive. It just wasn't a good way to go about things."

Highlight (Yellow) | Location 1321

This left Musk searching for an industry that had tons of money and inefficiencies that he and the Internet could exploit.

Highlight (Yellow) | Location 1333

He inquired as to how much Brazilian debt might be available at the 25-cents price. "The guy said, 'How much do you want?' and I came up with some ridiculous number like ten billion dollars," Musk said. When the trader confirmed that was doable, Musk hung up the phone. "I was thinking that they had to be fucking crazy because you could double your money. Everything was backed by Uncle Sam. It was a no-brainer."

Highlight (Yellow) | Location 1340

“I tried to tell them that’s not the point,” Musk said. “The point is that it’s fucking backed by Uncle Sam. It doesn’t matter what the South Americans do. You cannot lose unless you think the U.S. Treasury is going to default. But they still didn’t do it, and I was stunned. Later in life, as I competed against the banks, I would think back to this moment, and it gave me confidence. All the bankers did was copy what everyone else did. If everyone else ran off a bloody cliff, they’d run right off a cliff with them. If there was a giant pile of gold sitting in the middle of the room and nobody was picking it up, they wouldn’t pick it up, either.”

Highlight (Yellow) | Location 1475

The Confinity team led by Levchin favored moving toward open-source software like Linux, while Musk championed Microsoft’s data-center software as being more likely to keep productivity high. This squabble may sound silly to outsiders, but it was the equivalent of a religious war to the engineers, many of whom viewed Microsoft as a dated evil empire and Linux as the modern software of the people. Two months after the merger, Thiel resigned and Levchin threatened to walk out over the technology rift. Musk was left to run a fractured company.

Highlight (Yellow) | Location 1538

“He comes from the school of thought in the public relations world that you let no inaccuracy go uncorrected,” said Vince Solitto, the former communications chief at PayPal. “It sets a precedent, and you should fight every out-of-place comma tooth and nail. He takes things very personally and usually seeks war.” The stronger critique of Musk during this period of his life was that he had succeeded to a large degree despite himself. Musk’s traits as a confrontational know-it-all and his abundant ego created deep, lasting fractures within his companies. While Musk consciously tried to temper his behavior, these efforts were not enough to win over investors and more experienced executives. At both Zip2 and PayPal, the companies’ boards came to the conclusion that Musk was not yet CEO material. It can also be argued that Musk had become a hyperbolic huckster, who overreached and oversold his companies’ technology.

Highlight (Yellow) | Location 1551

just as people became comfortable with buying things from Amazon.com and eBay, Musk made the great leap forward to full-fledged Internet banking. He would bring standard financial instruments online and then modernize the industry with a host of new concepts. He exhibited a deep insight into human nature that helped his companies pull off exceptional marketing, technology, and financial feats. Musk was already playing the entrepreneur game at the highest level and working the press and investors like few others could. Did he hype things up and rub people the wrong way? Absolutely—and with spectacular results.

Highlight (Yellow) | Location 1782

watching Star Trek at his house. He also tinkered. Walking to grade school one day, Mueller discovered a smashed clock in an alley and turned it into a pet project. Each day, he fixed some part of the clock—a gear, a spring—until he got it working. A similar thing happened with the family’s lawn mower, which Mueller

disassembled one afternoon on the front lawn for fun. “My dad came home and was so mad because he thought he’d have to buy a new mower,” Mueller said. “But I put it back together, and it ran.” Mueller then got stuck on rockets. He started buying mail order kits and following the instructions to build small rockets.

Highlight (Yellow) | Location 1800

His crowning achievement was an eighty-pound engine that could produce thirteen thousand pounds of thrust and earned accolades as the world’s largest liquid-fuel rocket engine built by an amateur. “I still keep the rockets hanging in my garage,” Mueller said.

Highlight (Yellow) | Location 1850

SpaceX was to be America’s attempt at a clean slate in the rocket business, a modernized reset. Musk felt that the space industry had not really evolved in about fifty years. The aerospace companies had little competition and tended to make supremely expensive products that achieved maximum performance. They were building a Ferrari for every launch, when it was possible that a Honda Accord might do the trick. Musk, by contrast, would apply some of the start-up techniques he’d learned in Silicon Valley to run SpaceX lean and fast and capitalize on the huge advances in computing power and materials that had taken place over the past couple of decades.

Highlight (Yellow) | Location 1856

At a time when the cost of sending a 550-pound payload started at \$30 million, he promised that the Falcon 1 would be able to carry a 1,400-pound payload for \$6.9 million.

Highlight (Yellow) | Location 2154

Searching for a new site, Shotwell and Hans Koenigsmann put a Mercator projection of the world up on the wall and looked for a name they recognized along the equator, where the planet spins faster and gives rockets an added boost. The first name that jumped out was Kwajalein Island—or Kwaj—the largest island in an atoll between Guam and Hawaii in the Pacific Ocean and part of the Republic of the Marshall Islands. This spot registered with Shotwell because the U.S. Army had used it for decades as a missile test site. Shotwell looked up the name of a colonel at the test site and sent him an e-mail, and three weeks later got a call back with the army saying they would love to have SpaceX fly from the islands. In June 2005, SpaceX’s engineers began to fill containers with their equipment to ship them to Kwaj.

Highlight (Yellow) | Location 2820

One employee missed an event to witness the birth of his child. Musk fired off an e-mail saying, “That is no excuse. I am extremely disappointed. You need to figure out where your priorities are. We’re changing the world and changing history, and you either commit or you don’t.”*

Highlight (Yellow) | Location 2822

Marketing people who made grammatical mistakes in e-mails were let go, as were other people who hadn’t done anything “awesome” in recent memory. “He can be incredibly intimidating at times but doesn’t have a real sense for just how imposing he can be,” said one former Tesla executive.

Highlight (Yellow) | Location 2825

If you told him that you made a particular choice because ‘it was the standard way things had always been done,’ he’d kick you out of a meeting fast. He’d say, ‘I never want to hear that phrase again. What we have to do is fucking hard and half-assing things won’t be tolerated.’ He just destroys you and, if you survive, he determines if he can trust you. He has to understand that you’re as crazy as he is.”

Highlight (Yellow) | Location 2847

Lyons also saw an exhausted, stressed-out Musk spit coffee across a conference room table because it was cold and then, without a pause, demand that the employees work harder, do more, and mess up less. Like so many people privy to these performances, Lyons came away with no illusions about Musk’s personality but with the utmost respect for his vision and drive to execute. “Working at Tesla back then was like being Kurtz in *Apocalypse Now*,” Lyons said. “Don’t worry about the methods or if they’re unsound. Just get the job done. It comes from Elon. He listens, asks good questions, is fast on his feet, and gets to the bottom of things.”

Highlight (Yellow) | Location 2915

Musk likes to place one-dollar wagers on all manner of things—Can you catch herpes from the Great Barrier Reef? Is it possible to balance two forks with a toothpick?—that he knows he will win.

Highlight (Yellow) | Location 3086

On July 30, 2008, the Falcon 9 had a successful test fire in Texas with all nine of its engines lighting up and producing 850,000 pounds of thrust.

Highlight (Yellow) | Location 3140

“The people watching back at the factory were trying their best not to throw up,” said James McLaury, a metrology manager at SpaceX. Despite their past flubs, the engineers on Kwaj were confident that this launch would work. Some of these people had spent years on the island going through one of the more surreal engineering exercises in human history. They had been separated from their families, assaulted by the heat, and exiled on their tiny launchpad outpost—sometimes without much food—for days on end as they waited for the launch windows to open and dealt with the aborts that followed. So much of that pain and suffering and fear would be forgotten if this launch went successfully.

Highlight (Yellow) | Location 3249

Musk bluffed. He told the investors that he would take another loan from SpaceX and fund the entire round—all \$40 million—himself. The tactic worked. “When you have scarcity, it naturally reinforces greed and leads to more interest,”

Highlight (Yellow) | Location 3269

“Most people who are under that sort of pressure fray,” Gracias said. “Their decisions go bad. Elon gets hyperrational. He’s still able to make very clear, long-term decisions. The harder it gets, the better he gets. Anyone who saw what he went through firsthand came away with more respect for the guy. I’ve just never seen anything like his ability to take pain.”

Highlight (Yellow) | Location 3276

THE FALCON 9 HAS BECOME SPACEX’S WORKHORSE. The rocket looks—let’s face it—like a giant white phallus. It stands 224.4 feet tall, is 12 feet across, and weighs 1.1 million pounds. The rocket is powered by nine engines arranged in an “octaweb” pattern at its base with one engine in the center and eight others encircling it. The engines connect to the first stage, or the main body of the rocket, which bears the blue SpaceX insignia and an American flag. The shorter second stage of the rocket sits on top of the first and is the one that actually ends up doing things in space. It can be outfitted with a rounded container for carrying satellites or a capsule capable of transporting humans.

Highlight (Yellow) | Location 3287

About four hours before a launch, the Falcon 9 starts getting filled with an immense amount of liquid oxygen and rocket-grade kerosene. Some of the liquid oxygen vents out of the rocket as it awaits launch and is kept so cold that it boils off on contact with the metal and air, forming white plumes that stream down the rocket’s sides. This gives the impression of the Falcon 9 huffing and puffing as it limbers up before the journey.

Highlight (Yellow) | Location 3299

Seeing something so large hold so straight and steady while suspended in midair is hard for the brain to register. It is foreign, inexplicable.

Highlight (Yellow) | Location 3304

For Elon Musk, this spectacle has turned into a familiar experience. SpaceX has metamorphosed from the joke of the aeronautics industry into one of its most consistent operators. SpaceX sends a rocket up about once a month, carrying satellites for companies and nations and supplies to the International Space Station. Where the Falcon 1 blasting off from Kwajalein was the work of a start-up, the Falcon 9 taking off from Vandenberg is the work of an aerospace superpower. SpaceX can undercut its U.S. competitors—Boeing, Lockheed Martin, Orbital Sciences—on price by a ridiculous margin.

Highlight (Yellow) | Location 3309

Where these competitors rely on Russian and other foreign suppliers, SpaceX makes all of its machines from scratch in the United States. Because of its low costs, SpaceX has once again made the United States a player in the worldwide commercial launch market. Its \$60 million per launch cost is much less than what Europe and Japan charge and trumps even the relative bargains offered by the Russians and Chinese, who have the added benefit of decades of sunk government investment into their space programs as well as cheap labor.

Highlight (Yellow) | Location 3329

the retirement of the space shuttle made the United States totally dependent on the Russians to get astronauts to the ISS. Russia gets to charge \$70 million per person for the trip and to cut the United States off as it sees fit during political rifts. At present, SpaceX looks like the best hope of breaking this cycle and giving back to America its ability to take people into space.

Highlight (Yellow) | Location 3506

Musk initially relied on textbooks to form the bulk of his rocketry knowledge. But as SpaceX hired one brilliant person after another, Musk realized he could tap into their stores of knowledge. He would trap an engineer in the SpaceX factory and set to work grilling him about a type of valve or specialized material. “I thought at first that he was challenging me to see if I knew my stuff,” said Kevin Brogan, one of the early engineers. “Then I realized he was trying to learn things. He would quiz you until he learned ninety percent of what you know.” People who have spent significant time with Musk will attest to his abilities to absorb incredible quantities of information with near-flawless recall. It’s one of his most impressive and intimidating skills and seems to work just as well in the present day as it did when he was a child

Highlight (Yellow) | Location 3585

The SpaceX engineers found out how to make the PICA material less expensively and improved the underlying recipe so that Dragon—from day one—could withstand the heat of a reentry coming back from Mars.* The total cost for Dragon came in at \$300 million, which would be on the order of 10 to 30 times less than capsule projects built by other companies. “The metal comes in, we roll it out, weld it, and make things,” Davis said. “We build almost everything in-house. That is why the costs have come down.”

Highlight (Yellow) | Location 3612

SpaceX produced the communication computer in record time, and it ended up as the first system of its kind to pass NASA’s protocol tests on the first try. NASA officials were forced to say “cuckoo” over and over again during meetings—a small act of defiance SpaceX had planned all along to torture NASA.

Highlight (Yellow) | Location 3617

Musk had to sign off on every expenditure over \$10,000. “It was his money that we were spending, and he was keeping an eye on it, as he damn well should,” Watson said. “He made sure nothing stupid was happening.” Decisions were made quickly during weekly meetings, and the entire company bought into them. “It was amazing how fast people would adapt to what came out of those meetings,” Watson said. “The entire ship could turn ninety degrees instantly.

Highlight (Yellow) | Location 3624

The stuff I have seen him do in his head is crazy. He can get in discussions about flying a satellite and whether we can make the right orbit and deliver Dragon at the same time and solve all these equations in real time. It’s

amazing to watch the amount of knowledge he has accumulated over the years. I don't want to be the person who ever has to compete with Elon. You might as well leave the business and find something else fun to do. He will outmaneuver you, outthink you, and out-execute you.

Highlight (Yellow) | Location 3635

In one case someone spotted an error in a software file in the hours right before a launch. SpaceX's engineers changed the file, checked how it affected the test hardware, and, when no problems were detected, sent the file to the Falcon 9, waiting on the launchpad, all in less than thirty minutes. "NASA wasn't used to this," Watson said. "If something went wrong with the shuttle, everyone was just resigned to waiting three weeks before they could try and launch again."

Highlight (Yellow) | Location 3680

"There is a fundamental problem with regulators. If a regulator agrees to change a rule and something bad happens, they could easily lose their career. Whereas if they change a rule and something good happens, they don't even get a reward. So, it's very asymmetric. It's then very easy to understand why regulators resist changing the rules. It's because there's a big punishment on one side and no reward on the other. How would any rational person behave in such a scenario?"

Highlight (Yellow) | Location 3787

"If you hate people and think human extinction is okay, then fuck it," Shotwell said. "Don't go to space. If you think it is worth humans doing some risk management and finding a second place to go live, then you should be focused on this issue and willing to spend some money. I am pretty sure we will be selected by NASA to drop landers and rovers off on Mars. Then the first SpaceX mission will be to drop off a bunch of supplies, so that once people get there, there will be places to live and food to eat and stuff for them to do."

Highlight (Yellow) | Location 3895

This means that a machine guided by a computer formed the engine out of single piece of metal—in this case the high-strength alloy Inconel—so that its strength and performance should exceed anything built by humans by welding various parts together. And most mind-boggling of all, Musk revealed that the Dragon 2 will be able to land anywhere on Earth that SpaceX wants by using the SuperDraco engines and thrusters to come to a gentle stop on the ground. No more landings at sea. No more throwing spaceships away. "That is how a twenty-first-century spaceship should land," Musk said. "You can just reload propellant and fly again. So long as we continue to throw away rockets and spacecraft, we will never have true access to space."

Highlight (Yellow) | Location 3901

One of the company's next milestones will be the first flight of the Falcon Heavy, which is designed to be the world's most powerful rocket.* SpaceX has found a way to combine three Falcon 9s into a single craft with 27 of the Merlin engines and the ability to carry more than 53 metric tons of stuff into orbit.

Highlight (Yellow) | Location 3983

Most of the energy (about 70 percent) is lost as heat in the engine, while the rest is lost through wind resistance, braking, and other mechanical functions. The Model S, by contrast, has about a dozen moving parts, with the battery pack sending energy instantly to a watermelon-sized motor that turns the wheels. The Model S ends up being about 60 percent efficient, losing most of the rest of its energy to heat. The sedan gets the equivalent of about 100 miles per gallon.*

Highlight (Yellow) | Location 4008

Overnight, the Model S sometimes got new traction controls for hilly and highway driving or could suddenly recharge much faster than before or possess a new range of voice controls. Tesla had transformed the car into a gadget—a device that actually got better after you bought it. As Craig Venter, one of the earliest Model S owners and the famed scientist who first decoded man’s DNA, put it, “It changes everything about transportation. It’s a computer on wheels.”

Highlight (Yellow) | Location 4084

Model S, Musk and Tesla’s engineers knew from the start that they would place the 1,300-pound battery pack on the base of the car. This would give the vehicle a low center of gravity and excellent handling. It would also give the Model S what’s known as a low polar moment of inertia, which relates to how a car resists turning. Ideally, you want heavy parts like the engine as close as possible to the car’s center of gravity, which is why the engines of race cars tend to be near the middle of the vehicle. Traditional cars are a mess on this metric, with the bulky engine up front, passengers in the middle, and gasoline sloshing around the rear. In the case of the Model S, the bulk of the car’s mass is very close to the center of gravity and this has positive follow-on effects to handling, performance, and safety.

Highlight (Yellow) | Location 4376

You did what Musk asked or were prepared to burrow down into the properties of materials to explain why something could not be done. “He always said, ‘Take it down to the physics,’” Javidan said.

Highlight (Yellow) | Location 4748

Enough solar energy hits the Earth’s surface in about an hour to equal a year’s worth of worldwide energy consumption from all sources put together.²⁰

Highlight (Yellow) | Location 4825

\$35,000 price target for the Model 3 but also to pave the way for electric vehicles with 500-plus miles of range.

Highlight (Yellow) | Location 4850

Musk will build a second Gigafactory to compete with the Nevada site and place his own employees in competition with each other in a race to make the batteries first. “We’re not really trying to sort of yank anyone’s chain here,” Musk said. “It’s just like this thing needs to be completed on time.

Highlight (Yellow) | Location 4891

SpaceX will begin testing its ability to take people into space. It wants to perform a manned test flight by 2016 and to fly astronauts to the International Space Station for NASA the next year.

Highlight (Yellow) | Location 4913

At around \$1 million or \$500,000 per person, I think it's highly likely that there will be a self-sustaining Martian colony. There will be enough people interested who will sell their stuff on Earth and move. It's not about tourism. It's like people coming to America back in the New World days. You move, get a job there, and make things work. If you solve the transport problem, it's not that hard to make a pressurized transparent greenhouse to live in.

Highlight (Yellow) | Location 4917

"Eventually, you'd need to heat Mars up if you want it to be an Earthlike planet, and I don't have a plan for that. That would take a long time in the best of circumstances. It would probably take, I don't know, somewhere between a century and a millennium. There's zero chance of it being terraformed and Earthlike in my lifetime. Not zero, but 0.001 percent chance, and you would have to take real drastic measures with Mars."*

Highlight (Yellow) | Location 5054

When Musk announced in 2014 that Tesla would open-source all of its patents, analysts tried to decide whether this was a publicity stunt or if it hid an ulterior motive or a catch. But the decision was a straightforward one for Musk. He wants people to make and buy electric cars. Man's future, as he sees it, depends on this. If open-sourcing Tesla's patents means other companies can build electric cars more easily, then that is good for mankind, and the ideas should be free.

Highlight (Yellow) | Location 5066

"Elon has that deep appreciation for technology, the no-holds-barred attitude of a visionary, and that determination to go after long-term things that they both had," said Edward Jung, a child prodigy who worked for Jobs and Gates and ended up as Microsoft's chief software architect. "And he has that consumer sensibility of Steve along with the ability to hire good people outside of his own comfort areas that's more like Bill. You almost wish that Bill and Steve had a genetically engineered love child and, who knows, maybe we should genotype Elon to see if that's what happened."

Highlight (Yellow) | Location 5145

Tony Fadell, the former Apple executive credited with bringing the iPod and iPhone to market, has characterized the smartphone as representative of a type of super-cycle in which hardware and software have reached a critical point of maturity. Electronics are good and cheap, while software is more reliable and sophisticated. Their interplay is now resulting in science fiction-worthy ideas we were promised long ago becoming a reality.

Highlight (Yellow) | Location 5164

“Elon is one of the few people that I feel is more accomplished than I am,” said Craig Venter, the man who decoded the human genome and went on to create synthetic lifeforms. At some point he hopes to work with Musk on a type of DNA printer that could be sent to Mars. It would, in theory, allow humans to create medicines, food, and helpful microbes for early settlers of the planet. “I think biological teleportation is what is going to truly enable the colonization of space,” he said. “Elon and I have been talking about how this might play out.”

Highlight (Yellow) | Location 5168

One of Musk’s most ardent admirers is also one of his best friends: Larry Page, the cofounder and CEO of Google. Page has ended up on Musk’s house-surfing schedule. “He’s kind of homeless, which I think is sort of funny,” Page said. “He’ll e-mail and say, ‘I don’t know where to stay tonight. Can I come over?’”

Highlight (Yellow) | Location 5188

When Page and Sergey Brin began wondering aloud about developing ways to search the text inside of books, all of the experts they consulted said it would be impossible to digitize every book. The Google cofounders decided to run the numbers and see if it was actually physically possible to scan the books in a reasonable amount of time. They concluded it was, and Google has since scanned millions of books. “I’ve learned that your intuition about things you don’t know that much about isn’t very good,” Page said. “The way Elon talks about this is that you always need to start with the first principles of a problem. What are the physics of it? How much time will it take? How much will it cost? How much cheaper can I make it? There’s this level of engineering and physics that you need to make judgments about what’s possible and interesting. Elon is unusual in that he knows that, and he also knows business and organization and leadership and governmental issues.”

Highlight (Yellow) | Location 5234

“If you call someone a name, you get sent home. When I was going to school, if they punched you and there was no blood, it was like, ‘Whatever. Shake it off.’ Even if there was a little blood, but not a lot, it was fine.

Highlight (Yellow) | Location 5265

decided that this propensity for risk had little to do with Musk being insane, as he had wondered aloud several months earlier. No, Musk just seems to possess a level of conviction that is so intense and exceptional as to be off-putting to some.

Highlight (Yellow) | Location 5268

“I would like to die on Mars,” he said. “Just not on impact. Ideally I’d like to go for a visit, come back for a while, and then go there when I’m like seventy or something and then just stay there. If things turn out well, that would be the case. If my wife and I have a bunch of kids, she would probably stay with them on Earth.”

Highlight (Yellow) | Location 5279

plan to surround the Earth with thousands of small communications satellites. Musk wants, in effect, to build a space-based Internet in which the satellites would be close enough to the planet to beam down bandwidth at high speeds.

Highlight (Yellow) | Location 5286

To fund part of this unbelievably ambitious project, SpaceX secured \$1 billion from Google and Fidelity.

Highlight (Yellow) | Location 5414

Almost no one understands how PayPal actually worked or why it took off when other payment systems before and after it didn't. Most of the people at PayPal don't understand this. The reason it worked was because the cost of transactions in PayPal was lower than any other system. And the reason the cost of transactions was lower is because we were able to do an increasing percentage of our transactions as ACH, or automated clearinghouse, electronic transactions, and most importantly, internal transactions. Internal transactions were essentially fraud-free and cost us nothing. An ACH transaction costs, I don't know, like twenty cents or something. But it was slow, so that was the bad thing. It's dependent on the bank's batch processing time. And then the credit card transaction was fast, but expensive in terms of the credit card processing fees and very prone to fraud. That's the problem Square is having now.

Highlight (Yellow) | Location 5420

The critical thing is to achieve internal transactions. This is vital because they are instant, fraud-free, and fee-free. If you're a seller and have various options, and PayPal has the lowest fees and is the most secure, it's obviously the right thing to use.
